

Entropy and Gibbs Free Energy

 ΔS , $\Delta G = \Delta H - T\Delta S$ and K

Name _____ Class _____ Date _____

EQUATIONS AND DATA (298.15 K)

$$\Delta S^\circ = \sum nS^\circ(\text{products}) - \sum nS^\circ(\text{reactants}) \cdot \Delta G = \Delta H - T\Delta S \text{ (T in K; } \Delta S \text{ in kJ/K)} \cdot \Delta G^\circ = -RT \ln K \cdot$$

$$\Delta G^\circ = -nFE^\circ$$

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1} \cdot F = 96\,485 \text{ C/mol}$$

$$S^\circ \text{ (J mol}^{-1} \text{ K}^{-1}\text{): CH}_4\text{(g) } 186.3 \cdot \text{O}_2\text{(g) } 205.2 \cdot \text{CO}_2\text{(g) } 213.8 \cdot \text{H}_2\text{O(l) } 70.0 \cdot \Delta H_f^\circ \text{ (kJ/mol): CH}_4\text{(g) } -74.6 \cdot \text{CO}_2\text{(g) } -393.5 \cdot \text{H}_2\text{O(l) } -285.8$$

Show units on every quantity in every step.

A. Understanding

- (a) What is entropy? (b) State the second law of thermodynamics. (c) Why can an endothermic process, such as ice melting at 25 °C, be spontaneous?

- Predict the sign of ΔS : (a) $\text{H}_2\text{O(l)} \longrightarrow \text{H}_2\text{O(g)}$ (b) $2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{SO}_3\text{(g)}$ (c) $\text{NaCl(s)} \longrightarrow \text{Na}^+\text{(aq)} + \text{Cl}^-\text{(aq)}$ (d) $\text{CaCO}_3\text{(s)} \longrightarrow \text{CaO(s)} + \text{CO}_2\text{(g)}$ (e) $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{NO(g)}$

3. For each reaction, say whether it is spontaneous at all temperatures, never, only at low T, or only at high T: (a) $\Delta H = -120 \text{ kJ}$, $\Delta S = +85 \text{ J/K}$ (b) $\Delta H = +45 \text{ kJ}$, $\Delta S = -30 \text{ J/K}$ (c) $\Delta H = -60 \text{ kJ}$, $\Delta S = -150 \text{ J/K}$ (d) $\Delta H = +90 \text{ kJ}$, $\Delta S = +200 \text{ J/K}$

B. Calculations

4. Calculate ΔS° for $\text{CH}_4(\text{g}) + 2 \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l})$ and explain its sign.

5. Calculate ΔH° and ΔG° at 298.15 K for the same reaction.

6. For melting ice, $\Delta H = +6.01 \text{ kJ/mol}$ and $\Delta S = +22.0 \text{ J mol}^{-1} \text{ K}^{-1}$. (a) At what temperature is $\Delta G = 0$? (b) Find ΔG at -10.0°C and at $+10.0^\circ \text{C}$, and explain what each result means.

7. For $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$, $\Delta H^\circ = +179.2 \text{ kJ}$ and $\Delta S^\circ = +160.2 \text{ J/K}$. Calculate ΔG° at 1200 K. Is the decomposition spontaneous?

8. A reaction has $\Delta G^\circ = +10.0 \text{ kJ}$ at 298.15 K. Calculate K, and say whether reactants or products are favoured.

C. Challenge

9. The Daniell cell has $E^\circ = 1.10 \text{ V}$ and $n = 2$. Calculate (a) ΔG° (b) K at 298.15 K. What does the size of K tell you?